

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 638 Natural Language Processing — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Jurafsky & Martin — Speech and Language Processing, 3rd edition (free draft)	The primary text, free in full from the authors, and the best book in the field. It covers the classical and the neural material with equal care, which is unusual and is exactly what this course needs.
Eisenstein — Introduction to Natural Language Processing (free draft)	The rigorous companion. Stronger on the learning theory and the structured prediction material, and free as a draft PDF.
Goldberg — Neural Network Methods for NLP	Modules 04 through 06. Written before the transformer took over and consequently much clearer about <i>why</i> each architectural choice was made. Library copy.
The Illustrated Transformer, and The Annotated Transformer (free)	Module 06 twice over — once as a diagram and once as working code. Read the illustrated version first, then implement from the annotated one.
Bender et al. and the model-documentation literature (free)	Modules 12 and 13. The paper that framed the scale-and-harm argument, plus the model card and datasheet practices that came out of it.
Hugging Face course and documentation (free)	The practical toolchain for Modules 07 through 11 — free, current, and the implementations every exercise in the second half uses.

Module-by-module map

Module 01 — What Is Hard About Language

Resource	What it covers for this module
Jurafsky & Martin, chapters 1 and 2 (free)	§1 and §2 — the properties and the ambiguity levels, with far more examples.
Bender & Koller — Climbing towards NLU (free)	§3 and §4 argued sharply — what learning from form alone can and cannot provide. Read it again before Module 13.
Levesque et al. — The Winograd Schema Challenge (free)	§2's reference ambiguity as a designed test — and the reasoning about why it resists surface statistics.
Stanford CS224N, lecture 1 (free)	§1 as a lecture , with the representation argument made carefully.

Module 02 — Text as Data

Resource	What it covers for this module
Jurafsky & Martin, chapter 2 (free)	The whole module , with the normalisation decisions and the regular-expression machinery.
Sennrich et al. — Neural Machine Translation of Rare Words with Subword Units (free)	§2's algorithm in the original — short, and the motivation is stated plainly.
Kudo & Richardson — SentencePiece (free)	§2's variants , and the argument for removing language-specific pre-tokenisation entirely.
The Hugging Face tokenizers documentation (free)	§4's practical instruction , with the inspection API — use it on your own data today.

Module 03 — Language Models

Resource	What it covers for this module
Jurafsky & Martin, chapters 3 and 7 (free)	§1 through §4 , with the smoothing derivations and the perplexity discussion done properly.
Chen & Goodman — An Empirical Study of Smoothing Techniques (free)	§2's comparison, settled empirically — the study that established Kneser-Ney as the default.
Bengio et al. — A Neural Probabilistic Language Model (free)	§4 in the original , from 2003 — and the parameter sharing argument is made explicitly, which is why it is worth reading.
Jelinek — the history of statistical language modelling	§1 and §2 in context — how the field arrived at the next-token formulation, by somebody who was there.

Module 04 — Word Representations

Resource	What it covers for this module
Jurafsky & Martin, chapter 6 (free)	The whole module , including the PPMI weighting and the evaluation caveats of §2.

Resource	What it covers for this module
Mikolov et al., and Pennington et al. — word2vec and GloVe (free)	§1's objectives in the originals — both short, and worth reading for how modest the claims were.
Levy & Goldberg — Neural Word Embedding as Implicit Matrix Factorization (free)	§1's equivalence result — the paper that connected the two framings.
Bolukbasi et al., and Gonen & Goldberg (free)	§3's bias finding, and then the result that debiasing hides rather than removes it — read both, in that order.

Module 05 — Sequence Models

Resource	What it covers for this module
Jurafsky & Martin, chapter 9 (free)	§1 through §3 , with the LSTM equations written out carefully.
Olah — Understanding LSTM Networks (free)	§3 as the clearest available explanation — the diagrams are what make the gates click.
Pascanu et al. — On the difficulty of training RNNs (free)	§2's analysis in the original , including the clipping argument and the eigenvalue condition.
Hochreiter & Schmidhuber — Long Short-Term Memory (free)	§3 in the original , from 1997 — and the constant-error-carousel framing is the additive path argument stated plainly.

Module 06 — The Transformer

Resource	What it covers for this module
Vaswani et al. — Attention Is All You Need (free)	The original . Short, readable, and the ablations in it are more informative than the architecture diagram.
The Illustrated Transformer (free)	§1 through §3 visually — read this first if the equations are not landing.
The Annotated Transformer (free)	§4's implementation instruction, done for you — use it to check your own version rather than to replace writing one.
Michel et al. — Are Sixteen Heads Really Better than One? (free)	§2's prunability result , which is the honest caveat on head interpretation.

Module 07 — Pretraining and Transfer

Resource	What it covers for this module
Devlin et al., and Radford et al. — BERT and the GPT series (free)	§1's two objectives in the originals — read them together, because the contrast is the lesson.

Resource	What it covers for this module
Hoffmann et al. — Training Compute-Optimal Large Language Models (free)	§3's allocation correction — the result that established the earlier models were undertrained.
Tenney et al. — BERT Rediscovered the Classical NLP Pipeline (free)	§2's layer finding — and a satisfying connection to Module 01 §2's pipeline critique.
Schaeffer et al. — Are Emergent Abilities a Mirage? (free)	§3's sharpness caveat — the argument that the metric produces the discontinuity. Read with Module 10.

Module 08 — Adaptation

Resource	What it covers for this module
Hu et al. — LoRA (free)	§2 in the original — including the rank ablations, which are the evidence for the low-rank hypothesis.
Min et al. — Rethinking the Role of Demonstrations (free)	§3's format-versus-mapping result — the random-label experiment, which is the one to know.
Ouyang et al. — Training language models to follow instructions (free)	§4's two stages , with the preference pipeline described concretely.
Rafailov et al. — Direct Preference Optimization (free)	§4's simpler alternative — the same objective without the separate reward model.

Module 09 — Generation and Decoding

Resource	What it covers for this module
Holtzman et al. — The Curious Case of Neural Text Degeneration (free)	§2 in the original , and the paper that introduced top-p — the human-text-is-not-the-mode argument is made with data.
Jurafsky & Martin, the decoding sections (free)	§1's strategies , with beam search and length normalisation done carefully.
Welleck et al. — Neural Text Degeneration with Unlikelihood Training (free)	§2's repetition problem attacked at training time instead — a useful contrast with the decoding-time fixes.
The guidance and outlines libraries (free)	§3 as working code — grammar-constrained decoding you can use today.

Module 10 — Evaluation

Resource	What it covers for this module
Jurafsky & Martin, the evaluation sections (free)	§1 and §3 , with the metric definitions and their known failures.

Resource	What it covers for this module
Gururangan et al. — Annotation Artifacts in NLI Data (free)	§2's shortcut check, applied — the hypothesis-only baseline that reframed a whole benchmark.
Reiter — A Structured Review of the Validity of BLEU (free)	§3's first row, assessed carefully — where BLEU correlates with judgement and where it does not.
Zheng et al. — Judging LLM-as-a-Judge (free)	§3's callout with the measurements — the position, verbosity, and self-preference biases quantified.

Module 11 — Retrieval-Augmented Systems

Resource	What it covers for this module
Lewis et al. — Retrieval-Augmented Generation (free)	§1 in the original — and worth reading for how much of the current practice was already there.
Liu et al. — Lost in the Middle (free)	§4's fourth row, measured — the position effect across long contexts, which has direct design consequences.
Gao et al. — Retrieval-Augmented Generation: a survey (free)	§2 and §3's design space , organised — useful for finding the variant that matches your constraints.
Rashkin et al. — Measuring Attribution (free)	§4's last evaluation — a framework for whether a citation actually supports a claim.

Module 12 — Bias, Failure, and Fabrication

Resource	What it covers for this module
Bender et al. — On the Dangers of Stochastic Parrots (free)	§1 and §2's argument , and the paper that set the terms of this discussion. Read it in full.
Blodgett et al. — Language (Technology) is Power (free)	§1's callout, argued carefully — a survey of how 'bias' is used inconsistently and what that costs.
Ji et al. — Survey of Hallucination in NLG (free)	§3's mechanisms and mitigations , organised — and honest about how partial the mitigations are.
Mitchell et al. — Model Cards for Model Reporting (free)	§4's documentation practice — the template that made disaggregated reporting a norm rather than an exception.

Module 13 — Claiming Language Understanding Honestly

Resource	What it covers for this module
Bender & Koller — Climbing towards NLU (free)	§3's question, argued from the sceptical side — and worth rereading now that you have Module 07's evidence.

Resource	What it covers for this module
Rogers et al. — A Primer in BERTology (free)	§3's probing evidence, surveyed — what has actually been found in the representations, with the caveats.
Turpin et al. — Language Models Don't Always Say What They Think (free)	§3's load-bearing test, applied — stated reasoning that does not determine the answer.
Raji et al. — AI and the Everything in the Whole Wide World Benchmark (free)	§1's argument — why a benchmark cannot establish a general capability, developed properly.